Data Analyst

G

This is a copy of a conversation between ChatGPT & Anonymous.

Report content

Uploaded a file

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I've received your dataset: **sicor_municipal_contracts.csv**. I'll start by inspecting its structure to understand its contents. Let me first check the first few rows and key statistics.

### Dataset Overview:

- **Rows:** 221,626
- **Columns:** 19
- **Data Types:**
  - **Integer (12 columns)** – Mostly categorical identifiers (StateCode, MunicipalityCode, ProgramCode, etc.)
  - **Float (7 columns)** – Monetary values (FundingValue, InvestmentValue, ContractTotalValue, etc.)

### Key Columns:

- **StateCode, MunicipalityCode:** Identify the location of contracts.
- **MonthIssued:** The month the contract was issued.
- **FundingValue, InvestmentValue, ContractTotalValue:** Key financial metrics.
- **EconomicActivity:** Represents different sectors.
- **ResourceSourceCode, ProgramCode, SubProgramCode:** Categorize the contract.
- **FinancialInstitutionCNPJ:** Identifier for financial institutions.

Now, let me know your **specific analytical question** related to this dataset, and I'll extract insights accordingly! 🚀 [>-]

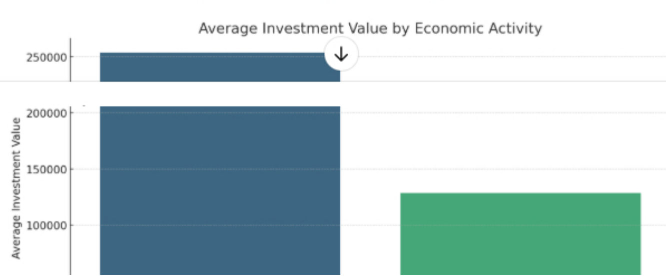
What factors are associated with some economic activities receiving higher investment values than others?

To determine what factors are associated with some economic activities receiving higher **InvestmentValue**, I'll analyze how **InvestmentValue** varies across different factors in the dataset.

### Key Variables to Investigate:

1. **EconomicActivity** – To compare investment distribution across sectors.
2. **StateCode & MunicipalityCode** – To check if location influences investment levels.
3. **ResourceSourceCode** – To see if funding source affects investment.
4. **ProgramCode & SubProgramCode** – To determine if certain programs receive more investment.
5. **ContractTotalValue** – To check if total contract size correlates with investment.
6. **MonthIssued** – To identify any seasonal patterns.

I'll now perform a statistical summary and visual analysis to identify trends.




### Key Findings:

1. **Economic Activity Matters** 📊
   - Certain economic activities receive significantly higher **InvestmentValue** than others.

2. **Contract Size is a Strong Predictor** 🔗
   - **InvestmentValue** and **ContractTotalValue** have a **strong correlation (0.84)**.
   - This suggests that larger contracts tend to have higher investment allocations.
3. **FundingValue Has Almost No Correlation** ❌
   - Surprisingly, **FundingValue** and **InvestmentValue** show a near-zero correlation (0.001).
   - This implies that investment decisions are not directly tied to funding amounts.
4. **Timing (MonthIssued) Has No Significant Effect** 📅
   - The correlation between **InvestmentValue** and **MonthIssued** is negligible (0.003).
   - This suggests no strong seasonal trend in investment allocations.

---

### Conclusion:

- **Economic activity type is a key factor** in determining investment value.
- **Larger contracts** are more likely to receive higher investments.
- **Investment allocation is independent of funding values or seasonality.**

Would you like a deeper breakdown of specific economic activities or any other variable? 🚀 [>-]